

Plymouth University

MODULE CODE: ELEC345/346

TITLE OF PAPER: HIGH SPEED COMMUNICATIONS

TIME ALLOWED

CILE

DATE

TIME

FACULTY

SCIENCE AND TECHNOLOGY

SCHOOL

ACADEMIC YEAR

2020/ 2021

STAGE

FINAL

INSTRUCTIONS TO CANDIDATES:

Candidates should attempt all **FOUR** questions
Marks for parts of questions are shown where appropriate.
State all assumptions, show all workings.

Candidates are not permitted to look at the examination paper until instructed to do so.

Release to library? Yes

Summer Exam

Q1. This question is about propagation and link budgets.

A lecturer has various parts to build a point-to-point Wi-Fi link to allow them to connect directly to the University Wi-Fi from their house in Saltash. It is line of sight to the top of Smeaton building and a distance of 7km. There is a 1dB loss due cabling to the antenna on Smeaton roof that has a gain of 12dBi. The second antenna also has a gain of 12dBi and each receiver has a noise figure of 6dB. Each OFDM channel has a bandwidth of 312.5 kHz and transmit power of 1.6mW

a) For the system described

- (i) What sort of antenna are these and why would you not find them on a normal Wi-Fi router?
- (ii) Are the units correct for all the numerical values given?
- (iii) What does noise figure refer to and how is it found?

[5 marks]

b) Sketch the system and showing all workings, units and stating any assumptions calculate:

- (i) The path loss and noise for the first OFDM channel at 2.4GHz
- (ii) The received power and signal to noise ratio (SNR)
- (iii) Will this system work if the SNR threshold is **** and by what margin is over or under this?

[10 marks]

c) Show all parts of the system (with numerical examples) that could be improved/increased (and can't) to make it work if it does not or enhance it if it does. Why might this system prove rather unreliable?

[10 marks]

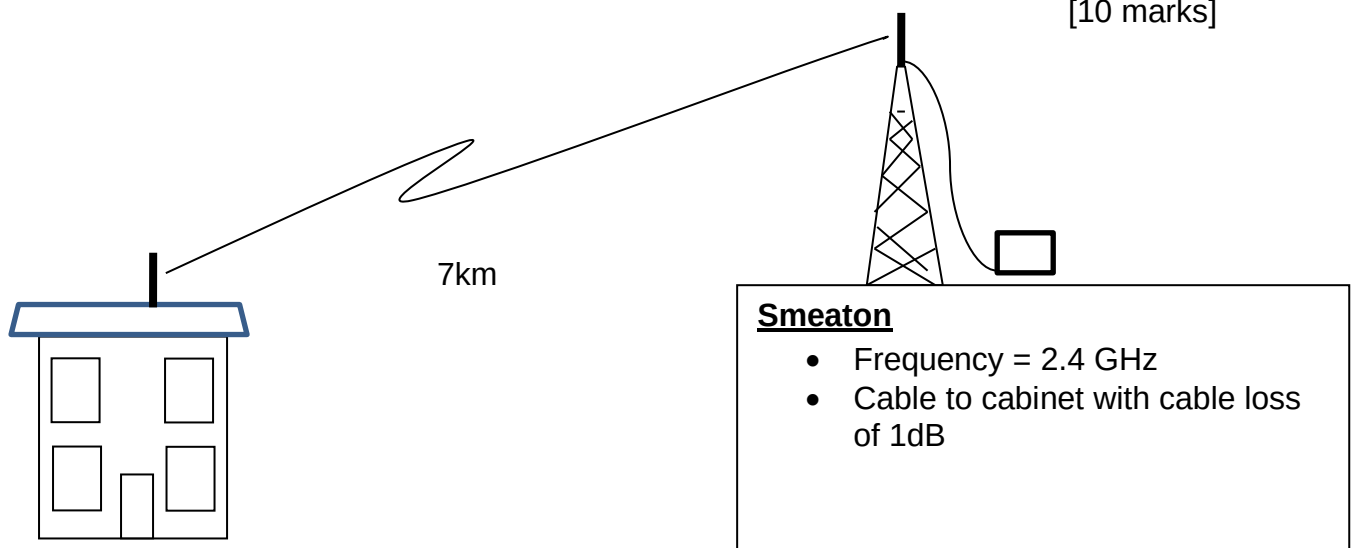


Figure 1 Campus to Saltash

(Over/....)

Q2. This question is about transmission lines and matching

- a) Given a 9-volt source with impedance $R_s = 25\Omega$ is connected to a transmission line with $Z_0 = 50\Omega$, length = 0.1m and $V_p = 1 \times 10^8$ m/s followed by a load $Z_L = 75\Omega$ calculate.
- The initial voltage on the line. [1 mark]
 - The voltage at the load after $5T$ (where T is the time taken to travel from one end of the line to the other), sketch a bounce diagram [6 marks]
 - The voltage the line will settle to. [1 mark]
 - When and why are transmission lines required? [5 marks]
- b) Given the PCB mounted 200MHz chip antenna shown in Figure 2 and a measured value of $20 + 70j$ design an L match. State all assumptions.

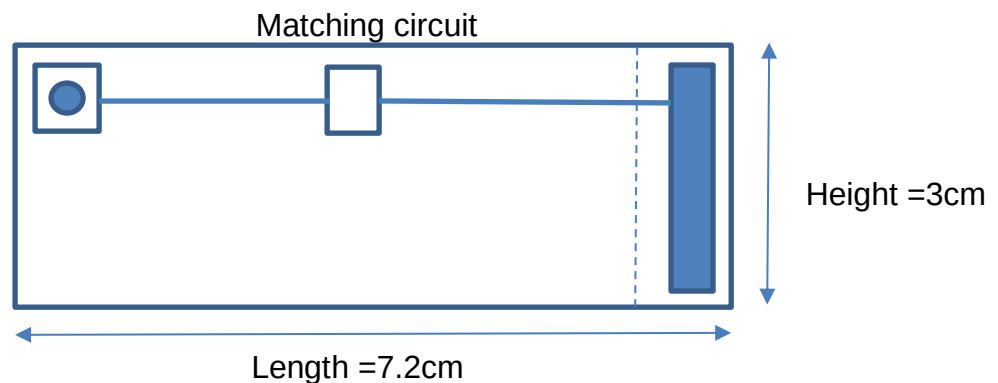


Figure 2 PCB chip antenna with matching area.

Q3. This question is about radio channel access.

- Explain with diagrams how Wi-Fi provides channel access in AD-HOC mode. What happens when the channel is busy or if there is a collision? Why might a collision occur and how might one be avoided? [9 marks]
- Explain the different methods of providing channel access in a cellular system including the uplink and downlink? [8 marks]
- Compare and contrast the difference between a smart antenna in a cellular system using beamforming and SDMA and a MIMO system in a Wi-Fi router. [8 marks]

(Over/....)

Q4. This question is on stub matching.

a) A load impedance, $Z_L = 30 + j30\Omega$ is connected to a 50Ω transmission line. Either by calculation or using the smith chart provided find:

- (i) The value on the chart
- (ii) The voltage reflection coefficient
- (iii) The voltage standing wave ratio in both linear form and dBs.
- (iv) Change to an admittance

[5 marks]

(b) For the load, $Y_L = 0.04 - j0.04\Omega$ use a parrallel stub to match the load to a 50Ω transmission line. Choose your terminations to keep all lengths as short as possible and state distances in wavelengths. How long would your physical stub be if $f=400\text{MHz}$?

[10 marks]

(c) Match in $Z_L = 30 + j30\Omega$ using a series, double stub for which the spacing between stubs is $3/8\lambda$ and the spacing between the first stub and the load is 0.05λ .

[10 marks]

END OF QUESTIONS